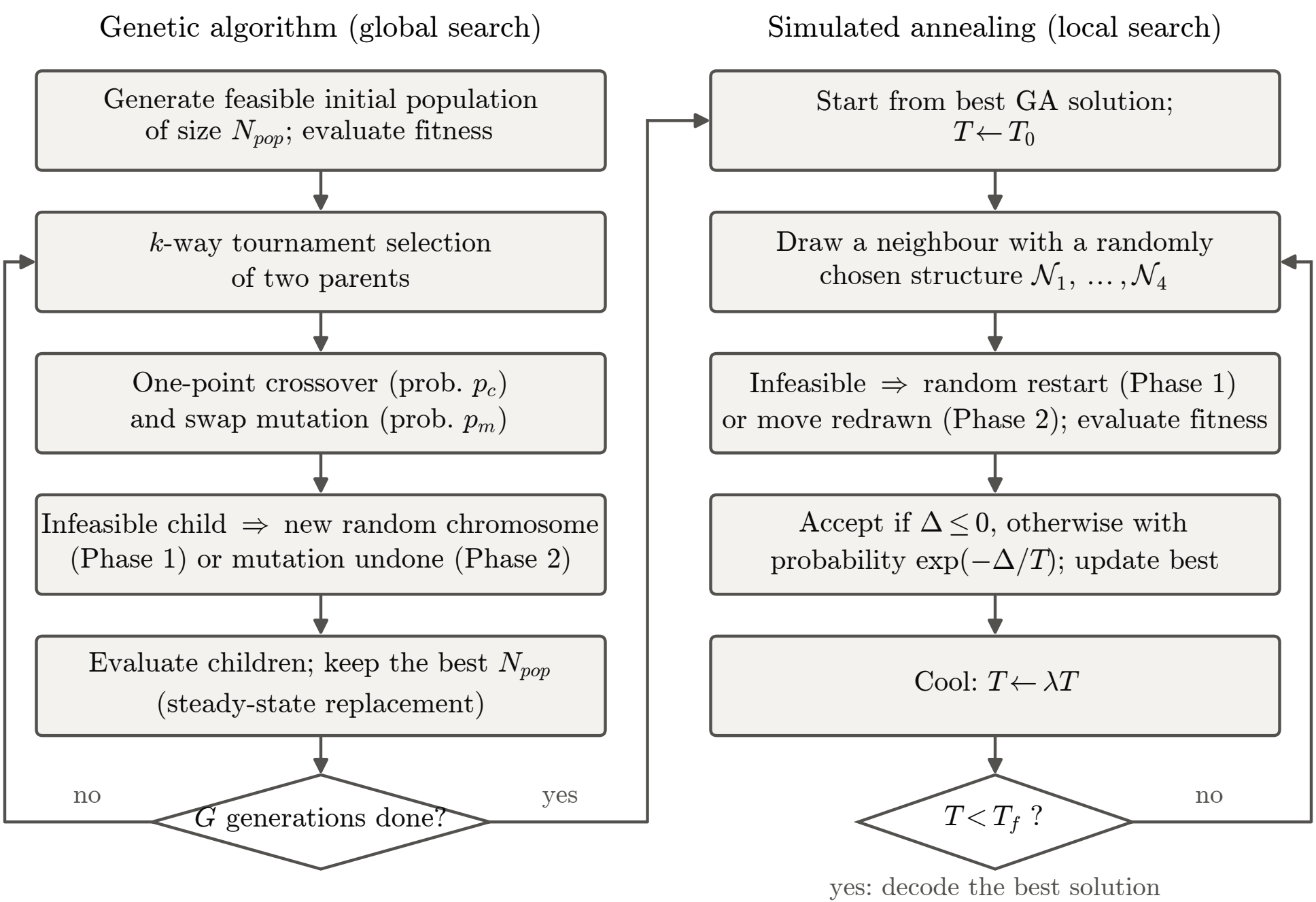
Genetic algorithm (global search)
Generate feasible initial population of size N_{pop}; evaluate fitness
k-way tournament selection of two parents
One-point crossover (prob. p_c) and swap mutation (prob. p_m)
Infeasible child $\Rightarrow$ new random chromosome (Phase 1) or mutation undone (Phase 2)
Evaluate children; keep the best N_{pop} (steady-state replacement)
G generations done?
no
yes
Simulated annealing (local search)
Start from best GA solution; $T \leftarrow T_0$
Draw a neighbour with a randomly chosen structure $\mathcal{N}_1, \ldots, \mathcal{N}_4$
Infeasible $\Rightarrow$ random restart (Phase 1) or move redrawn (Phase 2); evaluate fitness
Accept if $\Delta \leq 0$, otherwise with probability $\exp(-\Delta/T)$; update best
Cool: $T \leftarrow \lambda T$
$T < T_f$?
no
yes: decode the best solution